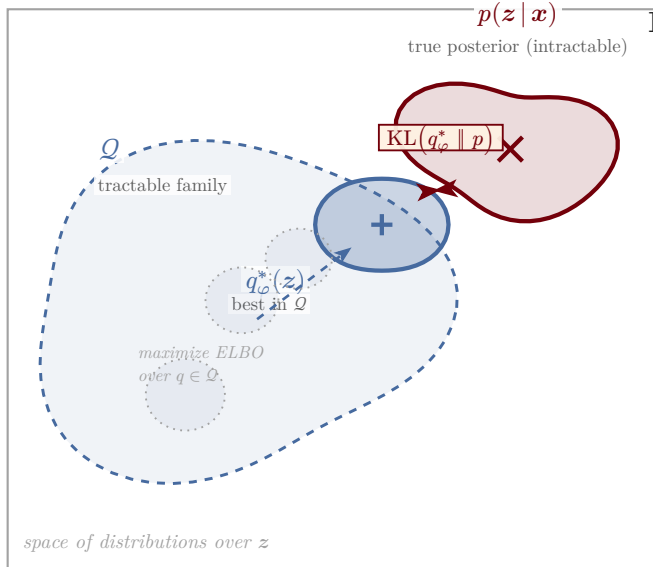
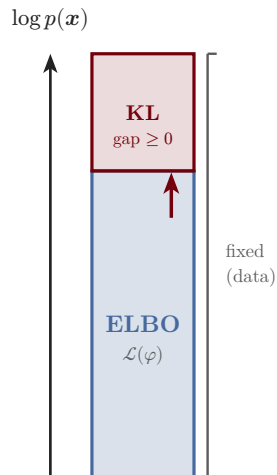


Variational inference: project $p(z \mid x)$ onto $\mathcal{Q}$



Evidence decomposition



$$\underbrace{\log p(x)}_{\text{log-evidence (fixed)}} = \underbrace{\mathcal{L}(\phi)}_{\text{ELBO}} + \underbrace{\text{KL}(q_\phi(z) \parallel p(z \mid x))}_{\geq 0}$$

$$\mathcal{L}(\phi) = \mathbb{E}_{q_\phi}[\log p(x, z)] - \mathbb{E}_{q_\phi}[\log q_\phi(z)] = \mathbb{E}_{q_\phi}[\log p(x, z)] + \mathcal{H}[q_\phi]$$

$$\arg \max_{q \in \mathcal{Q}} \mathcal{L}(\phi) \iff \arg \min_{q \in \mathcal{Q}} \text{KL}(q \parallel p(z \mid x))$$